

INTEGER SOLUTIONS OF $Z^2 - XY^2 = X^3 - 2$

ABSTRACT. We give an unconditional and self-contained classification of all integer triples satisfying $z^2 - xy^2 = x^3 - 2$. Elementary local arguments first force the positive integer x into a restricted set of residue classes and impose a splitting condition on every prime divisor of x . For square x , every solution is recovered from a finite divisor calculation; for nonsquare x , every solution is recovered uniquely from one of finitely many reduced seeds followed by a Pell recurrence. The proof includes an elementary existence proof for positive solutions of Pell's equation, so no external theorem is required.

1. STATEMENT OF THE CLASSIFICATION

We study

$$(1.1) \quad z^2 - xy^2 = x^3 - 2,$$

which is equivalently

$$(1.2) \quad z^2 + 2 = x(x^2 + y^2).$$

The equation is invariant under the independent sign changes $y \mapsto -y$ and $z \mapsto -z$. It is therefore natural to classify first the normalized solutions with $y, z \geq 0$, and then restore the signs.

Define the set of admissible positive integers

$$(1.3) \quad \mathcal{A} = \{x \in \mathbb{Z}_{>0} : x \text{ is odd, } x \not\equiv 2 \pmod{3}, p \mid x \text{ prime} \implies p \equiv 1 \text{ or } 3 \pmod{8}\}.$$

The term ‘‘admissible’’ refers only to necessary local conditions; membership in $\mathcal{A}$ is not sufficient for solubility.

For an odd integer $m \geq 3$, put

$$(1.4) \quad M_m = m^6 - 2$$

and define the finite set

$$(1.5) \quad \mathcal{F}_m = \left\{ a \in \mathbb{Z}_{>0} : a \mid M_m, \quad a \leq \frac{M_m}{a}, \quad \frac{M_m}{a} \equiv a \pmod{2m} \right\}.$$

For $a \in \mathcal{F}_m$, set

$$(1.6) \quad Y_{m,a} = \frac{M_m/a - a}{2m}, \quad Z_{m,a} = \frac{M_m/a + a}{2}.$$

Now let $x \geq 2$ be nonsquare. A positive Pell pair for x means a pair $(u, v) \in \mathbb{Z}_{>0}^2$ satisfying

$$(1.7) \quad u^2 - xv^2 = 1.$$

As will be proved in Section 3, such pairs exist. Among them there is a unique pair (u_x, v_x) for which

$$(1.8) \quad \varepsilon_x = u_x + v_x \sqrt{x}$$

assumes its least value. Put

$$(1.9) \quad N_x = x^3 - 2$$

and define the reduced seed set

$$(1.10) \quad \mathcal{S}_x = \{(r, s) \in \mathbb{Z}_{>0} \times \mathbb{Z}_{\geq 0} : r^2 - xs^2 = N_x, \quad s^2 < v_x^2 N_x\}.$$

This set is finite. For $(r, s) \in \mathcal{S}_x$ and $n \geq 0$, define $(Z_n, Y_n) = (Z_n(x; r, s), Y_n(x; r, s))$ by

$$(1.11) \quad Z_n + Y_n \sqrt{x} = (r + s \sqrt{x}) \varepsilon_x^n.$$

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Equivalently,

$$(1.12) \quad \begin{aligned} Z_0 &= r, & Y_0 &= s, \\ Z_{n+1} &= u_x Z_n + x v_x Y_n, & Y_{n+1} &= v_x Z_n + u_x Y_n. \end{aligned}$$

Theorem 1.1 (Complete classification). *Every integer solution of (1.1) has $x \in \mathcal{A}$, and all solutions are obtained as follows.*

(1) *The solutions with $x = 1$ are precisely*

$$(1.13) \quad (x, y, z) = (1, \pm 1, 0).$$

(2) *Let $x = m^2$, where $m \geq 3$ and $m^2 \in \mathcal{A}$. The solutions with this value of x are precisely*

$$(1.14) \quad (x, y, z) = (m^2, \sigma Y_{m,a}, \tau Z_{m,a}), \quad a \in \mathcal{F}_m, \quad \sigma, \tau \in \{\pm 1\}.$$

If $\mathcal{F}_m = \emptyset$, then this fiber has no solutions.

(3) *Let $x \in \mathcal{A}$ be nonsquare. The solutions with this value of x are precisely*

$$(1.15) \quad (x, y, z) = (x, \sigma Y_n(x; r, s), \tau Z_n(x; r, s)),$$

where $(r, s) \in \mathcal{S}_x$, $n \in \mathbb{Z}_{\geq 0}$, and $\sigma, \tau \in \{\pm 1\}$. If $\mathcal{S}_x = \emptyset$, then this fiber has no solutions.

After replacing y, z by $|y|, |z|$, the parameter a in the square case is unique, and the pair consisting of a reduced seed (r, s) and an exponent n in the nonsquare case is unique.

The proof occupies the next three sections.

2. LOCAL RESTRICTIONS ON x

We begin with the elementary quadratic-residue fact needed below. Its proof is included for completeness.

Lemma 2.1 (The residue class of primes representing minus two). *Let p be an odd prime. The congruence*

$$(2.1) \quad t^2 \equiv -2 \pmod{p}$$

has a solution if and only if

$$(2.2) \quad p \equiv 1 \text{ or } 3 \pmod{8}.$$

Proof. We give the standard elementary derivation. For $a \not\equiv 0 \pmod{p}$, write $\left(\frac{a}{p}\right) \in \{\pm 1\}$ for the Legendre symbol.

First, Fermat's congruence

$$(2.3) \quad a^{p-1} \equiv 1 \pmod{p}$$

follows because multiplication by a permutes the nonzero residue classes modulo p : multiplying all classes gives $a^{p-1}(p-1)! \equiv (p-1)! \pmod{p}$, and $(p-1)!$ is invertible modulo p . Define $\left(\frac{a}{p}\right) = 1$ when a is a nonzero square modulo p , and $\left(\frac{a}{p}\right) = -1$ otherwise. The nonzero squares modulo p are exactly the $(p-1)/2$ roots of $T^{(p-1)/2} - 1$: every square is a root by (2.3), the squares are pairwise distinct up to the identification $b \sim -b$, and a nonzero polynomial of degree d over a field has at most d roots, by induction from the factor theorem. If a is not a square, then $a^{(p-1)/2} \not\equiv 1 \pmod{p}$, while its square is 1 by (2.3); since p is odd, it must therefore be congruent to -1 . Consequently

$$(2.4) \quad a^{(p-1)/2} \equiv \left(\frac{a}{p}\right) \pmod{p}.$$

This is Euler's criterion. It also proves the multiplicativity of the Legendre symbol.

We next prove the form of Gauss's lemma that we need. Let $\nu(a)$ be the number of integers $j \in \{1, \dots, (p-1)/2\}$ for which the least positive residue of aj is greater than $p/2$. Replacing each such residue by its negative, the least absolute residues of $a, 2a, \dots, ((p-1)/2)a$ are, up to order, $1, 2, \dots, (p-1)/2$. Indeed, equality up to sign between two of them would give $a(i \mp j) \equiv 0$

(mod p), impossible for distinct $1 \leq i, j \leq (p-1)/2$. Multiplication and cancellation therefore give

$$(2.5) \quad a^{(p-1)/2} \equiv (-1)^{\nu(a)} \pmod{p}.$$

Together with (2.4), this is Gauss's lemma.

For $a = -1$, every one of the $(p-1)/2$ residues lies in the upper half, so

$$(2.6) \quad \left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}.$$

For $a = 2$, the residues $2j$, with $1 \leq j \leq (p-1)/2$, are all less than p , and $2j > p/2$ exactly when $j > p/4$. Thus

$$(2.7) \quad \nu(2) = \frac{p-1}{2} - \left\lfloor \frac{p}{4} \right\rfloor.$$

A direct check in the four odd residue classes modulo 8 yields

$$(2.8) \quad \left(\frac{2}{p}\right) = \begin{cases} 1, & p \equiv 1, 7 \pmod{8}, \\ -1, & p \equiv 3, 5 \pmod{8}. \end{cases}$$

Multiplicativity, (2.6), and (2.8) now give

$$(2.9) \quad \left(\frac{-2}{p}\right) = 1 \iff p \equiv 1 \text{ or } 3 \pmod{8},$$

which is equivalent to the asserted solvability criterion. $\square$

Proposition 2.2 (Necessary conditions on x). *If $(x, y, z) \in \mathbb{Z}^3$ satisfies (1.1), then $x \in \mathcal{A}$. In particular,*

$$(2.10) \quad x \equiv 1, 3, 9, \text{ or } 19 \pmod{24}.$$

Proof. Equation (1.2) shows first that $x > 0$. Indeed, its left-hand side is positive, whereas its right-hand side is zero when $x = 0$ and negative when $x < 0$.

Let p be an odd prime divisor of x . Reduction of (1.2) modulo p gives

$$(2.11) \quad z^2 \equiv -2 \pmod{p}.$$

By Lemma 2.1, every such p satisfies

$$(2.12) \quad p \equiv 1 \text{ or } 3 \pmod{8}.$$

We next prove that x is odd. Suppose to the contrary that x is even. Then (1.2) implies that z is even. Writing $z = 2w$, we have

$$(2.13) \quad z^2 + 2 = 2(2w^2 + 1),$$

so the exact exponent of 2 dividing $z^2 + 2$ is one. Consequently (1.2) forces

$$(2.14) \quad x = 2u$$

with u odd, and it also forces y to be odd: since x^2 is even, the factor $x^2 + y^2$ must be odd. Dividing (1.2) by 2 gives

$$(2.15) \quad 2w^2 + 1 = u(4u^2 + y^2).$$

Because u and y are odd,

$$(2.16) \quad 4u^2 + y^2 \equiv 5 \pmod{8}.$$

The left-hand side of (2.15) is congruent to 1 modulo 8 when w is even and to 3 modulo 8 when w is odd. Hence

$$(2.17) \quad 5u \equiv 1 \text{ or } 3 \pmod{8},$$

which gives

$$(2.18) \quad u \equiv 5 \text{ or } 7 \pmod{8}.$$

On the other hand, every prime divisor of u is an odd prime divisor of x , so it is congruent to 1 or 3 modulo 8 by (2.12). The set $\{1, 3\}$ is closed under multiplication modulo 8, because $3^2 \equiv 1 \pmod{8}$. It follows that $u \equiv 1$ or $3 \pmod{8}$, contradicting (2.18). Thus x is odd.

Finally, suppose that $x \equiv 2 \pmod{3}$. Reducing (1.1) modulo 3 gives

$$(2.19) \quad z^2 + y^2 \equiv 0 \pmod{3}.$$

As the only quadratic residues modulo 3 are 0 and 1, this forces $3 \mid y$ and $3 \mid z$. Thus the left-hand side of $z^2 - xy^2 = x^3 - 2$ is divisible by 9. But every integer congruent to 2 modulo 3 has cube congruent to 8 modulo 9, so

$$(2.20) \quad x^3 - 2 \equiv 6 \pmod{9},$$

a contradiction. Therefore $x \not\equiv 2 \pmod{3}$, and all conditions defining $\mathcal{A}$ hold.

Since x is odd and all its prime divisors are 1 or 3 modulo 8, one has $x \equiv 1$ or $3 \pmod{8}$. Combining this with $x \equiv 0$ or $1 \pmod{3}$ gives exactly the four classes in (2.10). $\square$

3. A FINITE-SEED THEOREM FOR GENERALIZED PELL EQUATIONS

We now develop, from first principles, the Pell machinery used in the nonsquare fibers.

Lemma 3.1 (Elementary rational approximation). *Let α be irrational. For every integer $Q \geq 1$, there exist integers p, q such that*

$$(3.1) \quad 1 \leq q \leq Q, \quad |q\alpha - p| < \frac{1}{Q} \leq \frac{1}{q}.$$

As Q varies, the resulting inequalities hold for infinitely many distinct positive integers q .

Proof. Consider the $Q + 1$ fractional parts

$$(3.2) \quad \{0\alpha\}, \{1\alpha\}, \dots, \{Q\alpha\}$$

in the Q half-open intervals $[j/Q, (j+1)/Q)$, $0 \leq j < Q$. Two fractional parts lie in the same interval. Subtracting them gives integers p, q , with $1 \leq q \leq Q$, satisfying $|q\alpha - p| < 1/Q$. The inequality $1/q \leq 1/Q$ follows from $q \leq Q$.

If only finitely many values of q occurred, then the nonzero numbers $|q\alpha - p|$ arising from those finitely many denominators would have a positive minimum. This contradicts the bound $|q\alpha - p| < 1/Q$ as $Q \rightarrow \infty$. $\square$

Theorem 3.2 (Existence of positive Pell solutions). *For every positive nonsquare integer D , there exist $u, v \in \mathbb{Z}_{>0}$ satisfying*

$$(3.3) \quad u^2 - Dv^2 = 1.$$

Proof. Apply Lemma 3.1 to $\alpha = \sqrt{D}$. Discarding finitely many pairs if necessary, we may assume $p, q > 0$. There are infinitely many distinct pairs (p, q) such that

$$(3.4) \quad |q\sqrt{D} - p| < \frac{1}{q}.$$

For each pair put

$$(3.5) \quad k = p^2 - Dq^2.$$

The integer k is nonzero because D is not a square. Moreover,

$$(3.6) \quad \begin{aligned} |k| &= |p - q\sqrt{D}|(p + q\sqrt{D}) \\ &< \frac{1}{q} \left(2q\sqrt{D} + \frac{1}{q} \right) < 2\sqrt{D} + 1. \end{aligned}$$

Thus k ranges over a finite set of nonzero integers. One value of k occurs for infinitely many pairs (p, q) . Among those pairs, two distinct pairs (p_1, q_1) and (p_2, q_2) are congruent componentwise modulo $|k|$, by the pigeonhole principle.

Define

$$(3.7) \quad U = \frac{p_1 p_2 - D q_1 q_2}{k}, \quad V = \frac{q_1 p_2 - p_1 q_2}{k}.$$

Both are integers. Indeed, modulo $|k|$,

$$(3.8) \quad p_1 p_2 - D q_1 q_2 \equiv p_1^2 - D q_1^2 = k \equiv 0,$$

and

$$(3.9) \quad q_1 p_2 - p_1 q_2 \equiv q_1 p_1 - p_1 q_1 = 0.$$

The identity

$$(3.10) \quad U + V\sqrt{D} = \frac{(p_1 + q_1\sqrt{D})(p_2 - q_2\sqrt{D})}{k}$$

then gives, after multiplication by its conjugate,

$$(3.11) \quad U^2 - DV^2 = \frac{(p_1^2 - Dq_1^2)(p_2^2 - Dq_2^2)}{k^2} = 1.$$

We have $V \neq 0$. Otherwise $p_1/q_1 = p_2/q_2$. Writing this common positive rational number in lowest terms as a/b , there would be positive integers t_1, t_2 with $(p_i, q_i) = t_i(a, b)$. Since both pairs have the same nonzero value k ,

$$(3.12) \quad t_1^2(a^2 - Db^2) = t_2^2(a^2 - Db^2),$$

so $t_1 = t_2$, contrary to the pairs being distinct. Equation (3.11) also implies $U \neq 0$. Therefore $(u, v) = (|U|, |V|)$ is a positive solution of (3.3). $\square$

Definition 3.3 (Least positive Pell unit). Let D be a positive nonsquare integer. Among all positive solutions of $u^2 - Dv^2 = 1$, choose the one minimizing $u + v\sqrt{D}$, and denote it by (u_D, v_D) . We write

$$(3.13) \quad \varepsilon_D = u_D + v_D\sqrt{D}.$$

This minimum exists: Theorem 3.2 supplies at least one positive solution, and below any fixed real bound there are only finitely many integer pairs (u, v) . Since $u_D^2 - Dv_D^2 = 1$, one has

$$(3.14) \quad \varepsilon_D^{-1} = u_D - v_D\sqrt{D} \in \mathbb{Z}[\sqrt{D}].$$

Theorem 3.4 (Finite-seed decomposition). Let $D, N \in \mathbb{Z}_{>0}$, with D nonsquare. Let $\varepsilon_D = u_D + v_D\sqrt{D}$ be as in Definition 3.3, and define

$$(3.15) \quad \mathcal{R}(D, N) = \{(r, s) \in \mathbb{Z}_{>0} \times \mathbb{Z}_{\geq 0} : r^2 - Ds^2 = N, \quad s^2 < v_D^2 N\}.$$

Then $\mathcal{R}(D, N)$ is finite. Every pair $(Z, Y) \in \mathbb{Z}_{>0} \times \mathbb{Z}_{\geq 0}$ satisfying

$$(3.16) \quad Z^2 - DY^2 = N$$

has a unique representation

$$(3.17) \quad Z + Y\sqrt{D} = (r + s\sqrt{D})\varepsilon_D^n$$

with $(r, s) \in \mathcal{R}(D, N)$ and $n \in \mathbb{Z}_{\geq 0}$. Conversely, every right-hand side of (3.17) has nonnegative integral coefficients and yields a solution of (3.16).

Proof. The set $\mathcal{R}(D, N)$ is finite because $0 \leq s < v_D\sqrt{N}$, and for each such integer s the equation $r^2 = N + Ds^2$ determines at most one positive integer r .

Let (Z, Y) satisfy (3.16), and put

$$(3.18) \quad \alpha = Z + Y\sqrt{D}.$$

Since $N > 0$, one has $Z^2 > DY^2$, hence $Z > Y\sqrt{D}$. Thus both α and its conjugate are positive, their product is N , and $\alpha \geq \sqrt{N}$. Because $\varepsilon_D > 1$, there is a unique integer $n \geq 0$ such that

$$(3.19) \quad \varepsilon_D^n \sqrt{N} \leq \alpha < \varepsilon_D^{n+1} \sqrt{N}.$$

Set

$$(3.20) \quad \gamma = \alpha \varepsilon_D^{-n}.$$

By (3.14), $\gamma \in \mathbb{Z}[\sqrt{D}]$, so

$$(3.21) \quad \gamma = r + s\sqrt{D}$$

for integers r, s . Its norm is N , and (3.19) gives

$$(3.22) \quad \sqrt{N} \leq \gamma < \varepsilon_D \sqrt{N}.$$

The conjugate of γ is N/γ , so

$$(3.23) \quad r = \frac{1}{2} \left(\gamma + \frac{N}{\gamma} \right) > 0, \quad s = \frac{1}{2\sqrt{D}} \left(\gamma - \frac{N}{\gamma} \right) \geq 0.$$

It remains to translate the upper bound in (3.22). The function

$$(3.24) \quad f(t) = t - \frac{N}{t}$$

is strictly increasing for $t > 0$, because $f'(t) = 1 + N/t^2 > 0$. Hence

$$(3.25) \quad \begin{aligned} 2s\sqrt{D} &= \gamma - \frac{N}{\gamma} \\ &< \varepsilon_D \sqrt{N} - \frac{N}{\varepsilon_D \sqrt{N}} \\ &= \sqrt{N} (\varepsilon_D - \varepsilon_D^{-1}) = 2v_D \sqrt{DN}. \end{aligned}$$

Therefore $s < v_D \sqrt{N}$, equivalently $s^2 < v_D^2 N$, and $(r, s) \in \mathcal{R}(D, N)$. This proves existence of the representation.

Conversely, let $(r, s) \in \mathcal{R}(D, N)$, and put $\gamma = r + s\sqrt{D}$. Since $r^2 - Ds^2 = N > 0$, one has $r > s\sqrt{D}$, so the conjugate of γ is positive. Moreover $s \geq 0$, and hence $\gamma \geq \sqrt{N}$. The seed inequality gives

$$(3.26) \quad f(\gamma) = 2s\sqrt{D} < 2v_D \sqrt{DN} = \sqrt{N}(\varepsilon_D - \varepsilon_D^{-1}) = f(\varepsilon_D \sqrt{N}).$$

Since f is strictly increasing, it follows that

$$(3.27) \quad \sqrt{N} \leq \gamma < \varepsilon_D \sqrt{N}.$$

Thus every member of the stated seed set lies in the same half-open reduction interval used above. Multiplication by ε_D^n gives integral coefficients; more explicitly, if

$$(3.28) \quad Z_n + Y_n \sqrt{D} = (r + s\sqrt{D})\varepsilon_D^n,$$

then

$$(3.29) \quad \begin{aligned} Z_{n+1} &= u_D Z_n + Dv_D Y_n, \\ Y_{n+1} &= v_D Z_n + u_D Y_n, \end{aligned}$$

so induction gives $Z_n > 0$ and $Y_n \geq 0$. Norm multiplicativity gives

$$(3.30) \quad Z_n^2 - DY_n^2 = (r^2 - Ds^2)(u_D^2 - Dv_D^2)^n = N.$$

For uniqueness, (3.27) shows that any representation $\alpha = (r + s\sqrt{D})\varepsilon_D^n$ with a stated seed satisfies

$$(3.31) \quad \varepsilon_D^n \sqrt{N} \leq \alpha < \varepsilon_D^{n+1} \sqrt{N}.$$

Hence n is the unique integer determined by (3.19). Once n is fixed, $r + s\sqrt{D} = \alpha \varepsilon_D^{-n}$ is fixed. Thus both the seed and the exponent are unique. $\square$

Corollary 3.5. *For fixed positive nonsquare D and positive N , the equation $Z^2 - DY^2 = N$ has either no solution in $\mathbb{Z}_{>0} \times \mathbb{Z}_{\geq 0}$ or infinitely many such solutions.*

Proof. By Theorem 3.4, a solution exists if and only if the finite seed set is nonempty. From any seed, the quantities $(r + s\sqrt{D})\varepsilon_D^n$ are strictly increasing as n increases, and therefore give infinitely many distinct solutions. $\square$

4. PROOF OF THE CLASSIFICATION

Proof of Theorem 1.1. Proposition 2.2 proves that every solution has $x \in \mathcal{A}$. It remains to classify each possible fiber.

The fiber $x = 1$. Equation (1.1) becomes

$$(4.1) \quad z^2 - y^2 = -1,$$

so

$$(4.2) \quad (z - y)(z + y) = -1.$$

The only ordered factor pairs of -1 are $(1, -1)$ and $(-1, 1)$. Solving gives $z = 0$ and $y = \pm 1$, proving (1.13).

Square fibers. Suppose $x = m^2$, with $m \geq 3$, and write

$$(4.3) \quad Y = |y|, \quad Z = |z|.$$

Then

$$(4.4) \quad Z^2 - m^2 Y^2 = m^6 - 2 = M_m.$$

Since $M_m > 0$, one has $Z > mY$. Thus

$$(4.5) \quad a = Z - mY, \quad b = Z + mY$$

are positive integers satisfying

$$(4.6) \quad ab = M_m, \quad a \leq b, \quad b - a = 2mY.$$

Therefore $b = M_m/a$ and

$$(4.7) \quad \frac{M_m}{a} \equiv a \pmod{2m},$$

so $a \in \mathcal{F}_m$. Solving (4.5) for Y, Z yields exactly (1.6).

Conversely, let $a \in \mathcal{F}_m$, and put $b = M_m/a$. Because m is odd, M_m is odd; hence its positive divisors a, b are odd. Thus

$$(4.8) \quad Z = \frac{a + b}{2} \in \mathbb{Z}.$$

The congruence in (1.5) gives

$$(4.9) \quad Y = \frac{b - a}{2m} \in \mathbb{Z}_{\geq 0}.$$

Moreover,

$$(4.10) \quad Z - mY = a, \quad Z + mY = b,$$

and therefore

$$(4.11) \quad Z^2 - m^2 Y^2 = ab = M_m.$$

This proves the square-fiber parameterization and its completeness. The normalized factor $a = Z - mY$ is determined uniquely by (Y, Z) .

Nonsquare fibers. Let $x \in \mathcal{A}$ be nonsquare. Since $x \geq 2$, the integer $N_x = x^3 - 2$ is positive. With $Y = |y|$ and $Z = |z|$, equation (1.1) becomes

$$(4.12) \quad Z^2 - xY^2 = N_x.$$

Apply Theorem 3.4 with $D = x$ and $N = N_x$. Its seed set is precisely $\mathcal{S}_x$, and its representation is precisely (1.11), equivalently (1.12). It gives every normalized solution and proves the uniqueness of the seed and exponent. Restoring the independent signs of y and z completes the proof. $\square$

Corollary 4.1 (Structure of the fibers). *Every square fiber of (1.1) is finite. Every nonsquare fiber is either empty or infinite. The full equation has infinitely many integer solutions.*

Proof. The square assertion follows from the finiteness of the divisor set $\mathcal{F}_m$. The nonsquare assertion follows from Corollary 3.5. Infinitely many solutions occur already in the fiber $x = 3$, as shown explicitly below. $\square$

5. EXAMPLES AND EFFECTIVE USE OF THE THEOREM

The classification is effective for each fixed x . In a square fiber, one factors $m^6 - 2$ and checks the finite congruence in (1.5). In a nonsquare fiber, one searches for the least positive solution of $u^2 - xv^2 = 1$; existence is guaranteed by Theorem 3.2. One then checks the finitely many integers $s \geq 0$ satisfying $s^2 < v_x^2 N_x$, retaining those for which $N_x + xs^2$ is a square.

Example 5.1 (The complete fiber $x = 3$). Here

$$(5.1) \quad N_3 = 25, \quad \varepsilon_3 = 2 + \sqrt{3}.$$

Indeed, $2^2 - 3 \cdot 1^2 = 1$, and no smaller positive Pell pair is possible because $v \geq 1$ and $u^2 = 1 + 3v^2 \geq 4$. A reduced seed satisfies

$$(5.2) \quad r^2 - 3s^2 = 25, \quad 0 \leq s < 5.$$

Modulo 5, this gives $r^2 \equiv 3s^2 \pmod{5}$. Since the nonzero squares modulo 5 are 1 and 4, the number 3 is a nonsquare; hence $5 \mid s$, and then $5 \mid r$. The bound in (5.2) forces $s = 0$, and then $r = 5$. Thus all normalized solutions with $x = 3$ are

$$(5.3) \quad Z_n + Y_n \sqrt{3} = 5(2 + \sqrt{3})^n, \quad n \geq 0.$$

The first pairs (Y_n, Z_n) are

$$(5.4) \quad (0, 5), (5, 10), (20, 35), (75, 130), (280, 485), \dots$$

Independent signs give all integer solutions in this fiber.

Example 5.2 (The complete fiber $x = 9$). Here $m = 3$ and

$$(5.5) \quad M_3 = 3^6 - 2 = 727.$$

The integer 727 has no prime divisor at most $\sqrt{727} < 27$: it is not divisible by 2, 3, 5, 7, 11, 13, 17, 19, or 23. Hence it is prime. The only normalized factor pair is $(a, b) = (1, 727)$, and $727 \equiv 1 \pmod{6}$. Formula (1.6) gives

$$(5.6) \quad Y = \frac{727 - 1}{6} = 121, \quad Z = \frac{727 + 1}{2} = 364.$$

Therefore all solutions with $x = 9$ are

$$(5.7) \quad (9, \pm 121, \pm 364).$$

Example 5.3 (Admissibility is not sufficient). The prime 43 belongs to $\mathcal{A}$, because $43 \equiv 3 \pmod{8}$ and $43 \equiv 1 \pmod{3}$. Nevertheless the fiber $x = 43$ is empty. Indeed,

$$(5.8) \quad 43^3 - 2 = 79505 \equiv 5 \pmod{25}.$$

If a solution existed, reduction modulo 5 would give

$$(5.9) \quad z^2 - 3y^2 \equiv 0 \pmod{5}.$$

As 3 is a nonsquare modulo 5, this forces $5 \mid y$ and $5 \mid z$. The left-hand side $z^2 - 43y^2$ would then be divisible by 25, whereas the right-hand side is congruent to 5 $\pmod{25}$, a contradiction.